

SECURITISATION RISK ANALYTICS AND IFRS 9 RISK ARENA USING POWER BI

Project Title

Securitisation Risk Analytics, IFRS 9 Expected Credit Loss Modeling, Stress Testing and Investor Reporting Dashboard using Power BI

Submitted By

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Technology Stack

- Power BI
- DAX
- Power Query
- Excel
- IFRS 9 Risk Modeling

Project Objective

The objective of this project is to design and implement a comprehensive securitisation risk analytics platform that enables financial institutions and investors to evaluate portfolio performance, monitor delinquency trends, estimate expected credit losses, perform stress testing, analyze tranche structures, and generate investor reports through interactive Power BI dashboards.

Abstract

Securitisation is a financial process in which pools of loans are packaged and converted into marketable securities. Investors require accurate risk assessment and transparent reporting to evaluate the performance of these portfolios. This project develops a Power BI-based securitisation risk analytics solution using auto loan portfolio data.

The solution integrates delinquency monitoring, IFRS 9 Expected Credit Loss (ECL) modeling, vintage analysis, tranche cashflow waterfall analysis, stress testing frameworks, investor reporting dashboards, and a gamified securitisation risk arena. Multiple datasets including loan

portfolio information, DPD snapshots, dynamic loss records, and static pool vintage data are integrated into a unified analytical model.

The project demonstrates how modern business intelligence tools can be used to transform raw loan data into actionable insights for risk managers, investors, and regulatory stakeholders.

Problem Statement and Objectives

Problem Statement

Financial institutions face significant challenges in monitoring credit risk and predicting future losses within securitised loan portfolios. Traditional reporting systems often fail to provide timely insights regarding delinquency trends, expected losses, and investor exposure.

Key challenges include:

1. Monitoring delinquency migration across DPD buckets.
2. Estimating expected credit losses under IFRS 9.
3. Understanding vintage performance.
4. Evaluating tranche-level cashflows.
5. Performing stress testing under adverse economic conditions.
6. Providing transparent investor reporting.

Project Objectives

The primary objectives of this project are:

- Build a securitisation risk analytics architecture.
- Design a scalable Power BI data model.
- Implement IFRS 9 Expected Credit Loss calculations.
- Develop DPD risk monitoring dashboards.
- Create vintage performance analysis.
- Design tranche waterfall cashflow models.
- Build stress testing frameworks.
- Generate investor reporting dashboards.

- Simulate economic crisis scenarios through a Risk Arena.
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Data Sources and Architecture

The project uses four primary datasets:

1. Auto Loan Securitisation Data

Contains loan-level information including:

- Loan ID
- Borrower Details
- Origination Date
- Current Balance
- Interest Rate
- Loan Type
- Default Status

2. DPD Snapshot History

Provides delinquency information:

- Loan ID
- Snapshot Date
- DPD Bucket
- Current Balance
- Days Past Due

3. Dynamic Loss Data

Captures loss and recovery information:

- Defaulted Amount
- Recovery Amount
- Net Loss

- Recovery Rate

4. Static Pool Vintage Data

Used for cohort analysis:

- Vintage Year
- Origination Period
- Pool Balance
- Performance Metrics

Architecture Overview

Data Sources → Power Query → Data Model → DAX Measures →
Power BI Dashboards → Investor Reporting

This layered architecture ensures scalability and maintainability of the analytical solution.

Data Model and Relationships

The Power BI data model was designed using a star-schema approach.

Tables

1. Auto Loan Securitisation Data (Fact Table)
2. DPD Snapshot History
3. Dynamic Loss Data
4. Static Pool Vintage Data
5. Date Table

Relationships

Auto Loan ↔ DPD Snapshot

Relationship based on:

LoanID → LoanID

Purpose:

Track delinquency movement over time.

Auto Loan ↔ Dynamic Loss Data

Relationship based on:

LoanID → LoanID

Purpose:

Link portfolio balances with loss information.

Date Table ↔ Snapshot Dates

Relationship based on:

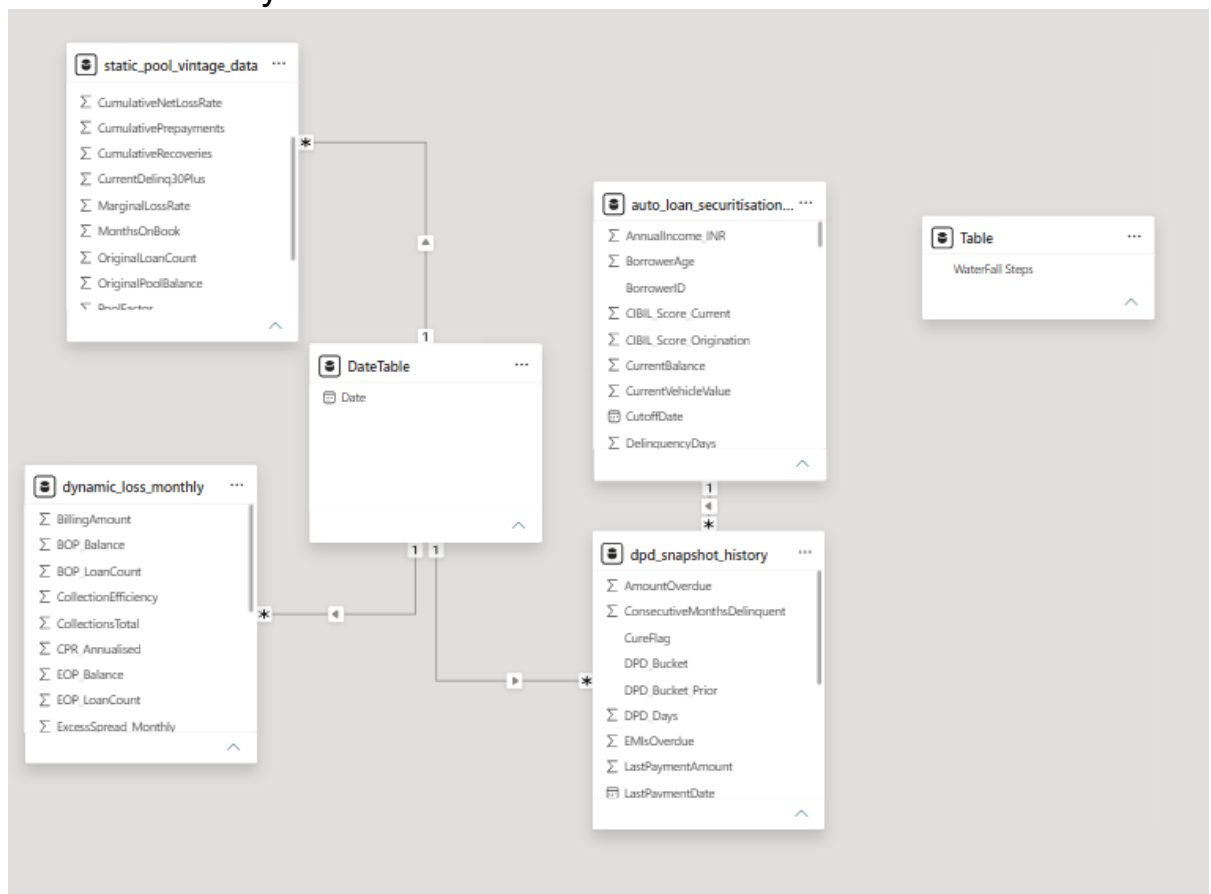
Date → SnapshotDate

Purpose:

Enable time intelligence calculations.

Benefits of the Model

- Improved query performance.
- Simplified DAX calculations.
- Enhanced dashboard responsiveness.
- Better scalability for future datasets



Executive Dashboard

Purpose

The Executive Dashboard serves as a high-level overview of the securitised portfolio. It enables management and investors to quickly assess portfolio performance, risk exposure, and key credit metrics.

Key Performance Indicators (KPIs)

Total Exposure

Represents the outstanding balance of all loans within the portfolio.

DAX Measure

Total Exposure =
SUM(auto_loan_securitisation_data[CurrentBalance])

Total Loans

Represents the total number of active loans.

DAX Measure

Total Loans =
COUNT(auto_loan_securitisation_data[LoanID])

Average Interest Rate

Shows the average lending rate across the portfolio.

DAX Measure

Average Interest Rate =
AVERAGE(auto_loan_securitisation_data[InterestRate])

Default Rate

Measures the proportion of defaulted loans.

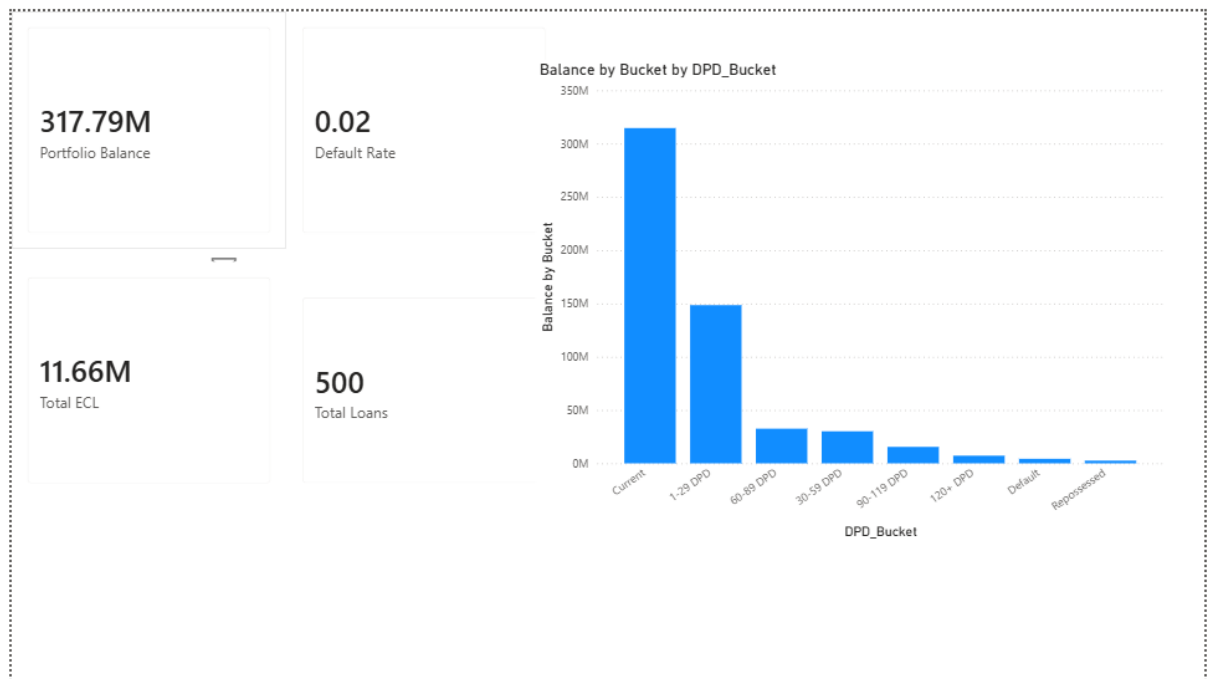
DAX Measure

Default Rate =
DIVIDE(
COUNTROWS(
FILTER(
auto_loan_securitisation_data,
auto_loan_securitisation_data[IsDefaulted]=TRUE()
)),

[Total Loans]
)

Dashboard Insights

- High exposure indicates concentration risk.
- Higher default rates increase expected losses.
- Interest rate trends influence borrower affordability.



DAX Measures and KPI Framework

Importance of DAX

Data Analysis Expressions (DAX) are used to perform calculations and derive business metrics.

Core Measures

Total Exposure

Total Exposure =
`SUM(auto_loan_securitisation_data[CurrentBalance])`

Probability of Default (PD)

PD =
DIVIDE(
SUMX(
auto_loan_securitisation_data,
IF(auto_loan_securitisation_data[IsDefaulted]=TRUE(),1,0)
),
COUNT(auto_loan_securitisation_data[LoanID])
)

Exposure at Default (EAD)

EAD =
[Total Exposure]

Loss Given Default (LGD)

Assumed at 40%.

LGD = 0.40

Expected Credit Loss

Expected Loss =
[PD] * [LGD] * [EAD]

Business Interpretation

These measures form the foundation of IFRS 9 risk calculations and help estimate future portfolio losses.

DPD Risk Analysis

Introduction

Days Past Due (DPD) is one of the most important indicators of credit risk.

DPD Buckets

Bucket Description

Current No delinquency

1-29 Early delinquency

30-59 Increased risk

60-89 High risk

90+ Default risk

Visualizations

DPD Distribution

Shows loan count across delinquency buckets.

Exposure by DPD

Shows balance concentration by bucket.

Roll Rate Analysis

Tracks migration between buckets over time.

Business Insights

- Rising 30+ DPD indicates worsening credit quality.
 - Concentration in 90+ bucket signals elevated losses.
 - Roll rates help forecast future defaults.
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DPD Analysis



IFRS 9 Expected Credit Loss Framework

Introduction

IFRS 9 requires banks and financial institutions to recognize expected losses before actual defaults occur.

Components

Probability of Default (PD)

Likelihood of borrower default.

Loss Given Default (LGD)

Percentage loss if default occurs.

Exposure at Default (EAD)

Outstanding balance at default.

Formula

Expected Credit Loss =
 $PD \times LGD \times EAD$

Benefits

- Forward-looking risk measurement
- Regulatory compliance
- Improved capital planning

Dashboard Outputs

- PD Card
 - LGD Card
 - EAD Card
 - Expected Loss Card
-

IFRS 9 Staging Methodology

Stage 1

Performing Loans

DPD < 30

Stage 2

Significant Increase in Credit Risk

$30 \leq \text{DPD} < 90$

Stage 3

Credit Impaired

DPD $\geq$ 90

DAX Column

IFRS Stage =

SWITCH(

TRUE(),

auto_loan_securitisation_data[DPD Days] < 30,"Stage 1",

auto_loan_securitisation_data[DPD Days] < 90,"Stage 2",

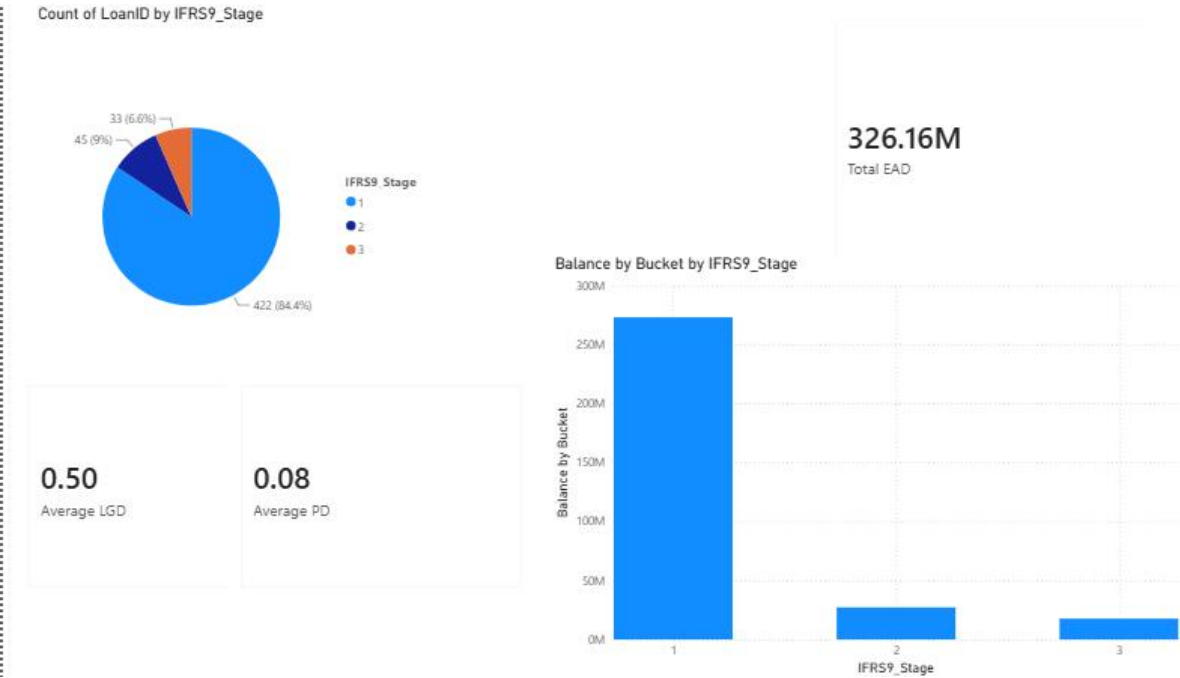
"Stage 3"

)

Business Significance

- Stage 1 loans require 12-month ECL.
- Stage 2 loans require lifetime ECL.
- Stage 3 loans are considered impaired.

IFRS9 Dashboard



Vintage Analysis

Introduction

Vintage Analysis is a cohort-based risk assessment technique used in securitisation and banking. It groups loans by origination period and tracks their performance over time.

The objective is to identify whether newer or older loan vintages exhibit higher delinquency and default rates.

Key Concepts

Vintage Year

The year in which the loan originated.

Example:

Loan ID	Origination Date	Vintage
L001	2021-05-12	2021
L002	2022-08-15	2022

MOB (Months on Book)

Represents how many months have passed since loan origination.

DAX Column

```
MOB =  
DATEDIFF(  
    auto_loan_securitisation_data[OriginationDate],  
    TODAY(),  
    MONTH  
)
```

Vintage Heatmap

Rows:

- Vintage Year

Columns:

- MOB

Values:

- Total Exposure

Conditional formatting is applied:

Green = Low Risk

Yellow = Medium Risk

Red = High Risk

Business Insights

- Older vintages generally exhibit higher default rates.
- New vintages indicate current underwriting quality.
- Investors use vintage analysis to evaluate pool performance.

Vintage Month	13-24	25-36	36+	Total
2021-01			13,10,707.18	13,10,707.18
2021-02			15,64,317.58	15,64,317.58
2021-03			40,33,601.89	40,33,601.89
2021-04			26,66,968.87	26,66,968.87
2021-05			41,02,057.73	41,02,057.73
2021-06			20,51,887.52	20,51,887.52
2021-07			36,60,515.25	36,60,515.25
2021-08			37,78,608.55	37,78,608.55
2021-09			56,09,039.12	56,09,039.12
2021-10			19,72,944.58	19,72,944.58
2021-11			25,60,146.82	25,60,146.82
2021-12			28,36,189.22	28,36,189.22
2022-01			38,38,640.07	38,38,640.07
2022-02			47,32,932.24	47,32,932.24
2022-03			65,26,764.00	65,26,764.00
2022-04			31,36,182.65	31,36,182.65
2022-05			31,66,853.51	31,66,853.51
2022-06			65,19,431.39	65,19,431.39
2022-07			52,54,164.28	52,54,164.28
2022-08			54,37,175.44	54,37,175.44
2022-09			65,05,074.90	65,05,074.90
2022-10			80,73,795.09	80,73,795.09
2022-11			74,39,739.00	74,39,739.00
2022-12			1,24,08,039.60	1,24,08,039.60
2023-01			58,08,161.13	58,08,161.13
2023-02			56,56,485.20	56,56,485.20
2023-03			60,99,490.94	60,99,490.94
Total	5,12,32,754.39	13,03,15,306.69	13,62,37,141.06	31,77,85,202.14

Waterfall and Tranche Analysis

Introduction

Securitisation converts loan cashflows into structured securities.

Cashflows are distributed according to a predefined waterfall hierarchy.

Tranche Structure

Senior Tranche

Lowest risk.

Receives payments first.

Mezzanine Tranche

Medium risk.

Receives payments after Senior Tranche.

Equity Tranche

Highest risk.

Absorbs losses first.

Net Cashflow

DAX Measure

Net Cashflow =
[Total Exposure] - [Expected Loss]

Tranche Allocation

Senior Tranche

Senior Tranche =
 $0.80 * [\text{Net Cashflow}]$

Mezzanine Tranche

Mezzanine Tranche =
 $0.15 * [\text{Net Cashflow}]$

Equity Tranche

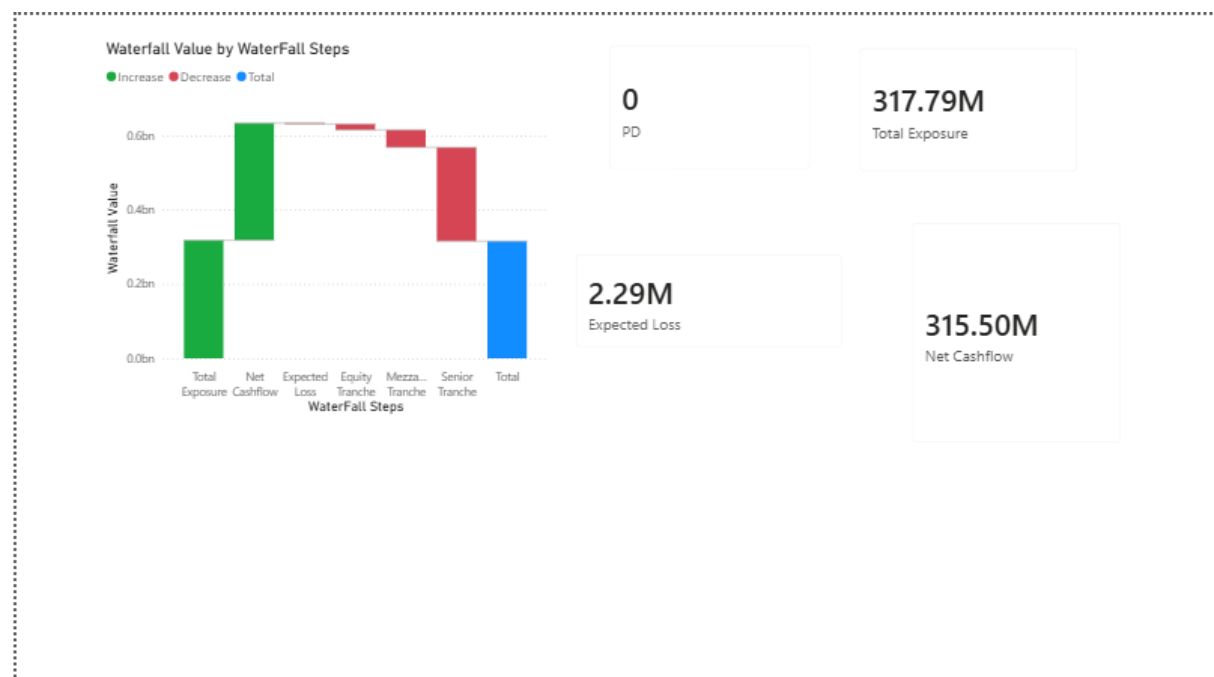
Equity Tranche =
 $0.05 * [\text{Net Cashflow}]$

Business Significance

Senior investors receive lower returns but higher security.

Equity investors receive higher returns but bear greater losses.

WaterFall Analysis



Stress Testing Framework

Introduction

Stress testing evaluates portfolio resilience under adverse economic scenarios.

The objective is to estimate losses during economic downturns.

Scenario 1: Base Case

Current economic conditions.

Base EL =
[Expected Loss]

Scenario 2: Mild Stress

Assumes moderate economic deterioration.

Mild Stress EL =
[Expected Loss] * 1.25

Scenario 3: Severe Stress

Assumes recession conditions.

Severe Stress EL =
[Expected Loss] * 1.50

Stress Testing Dashboard

KPIs:

- Base EL
- Mild Stress EL
- Severe Stress EL

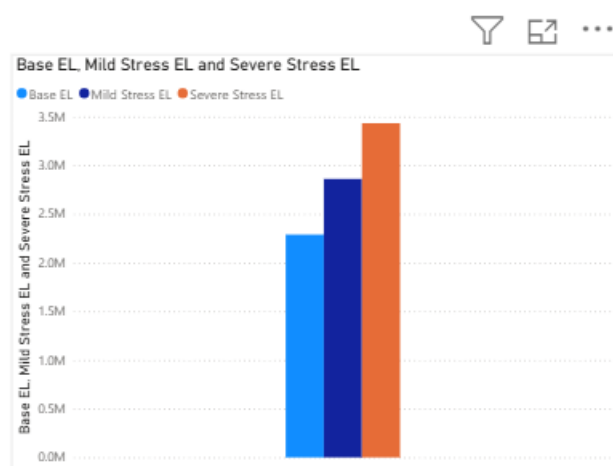
Visual:

Clustered Column Chart comparing scenarios.

Business Insights

- Losses increase significantly under stress.
- Severe scenarios highlight portfolio vulnerability.
- Supports risk mitigation planning.

StressTesting Framework



Investor Reporting Dashboard

Objective

Provide investors with transparency regarding portfolio performance and risk exposure.

Key Metrics

Total Exposure

Outstanding portfolio balance.

Probability of Default

Expected default frequency.

Expected Loss

Projected credit loss.

Net Cashflow

Expected investor cashflows.

Visual Components

Exposure by Loan Type

Donut Chart

Exposure by DPD Bucket

Column Chart

Waterfall Summary

Waterfall Visual

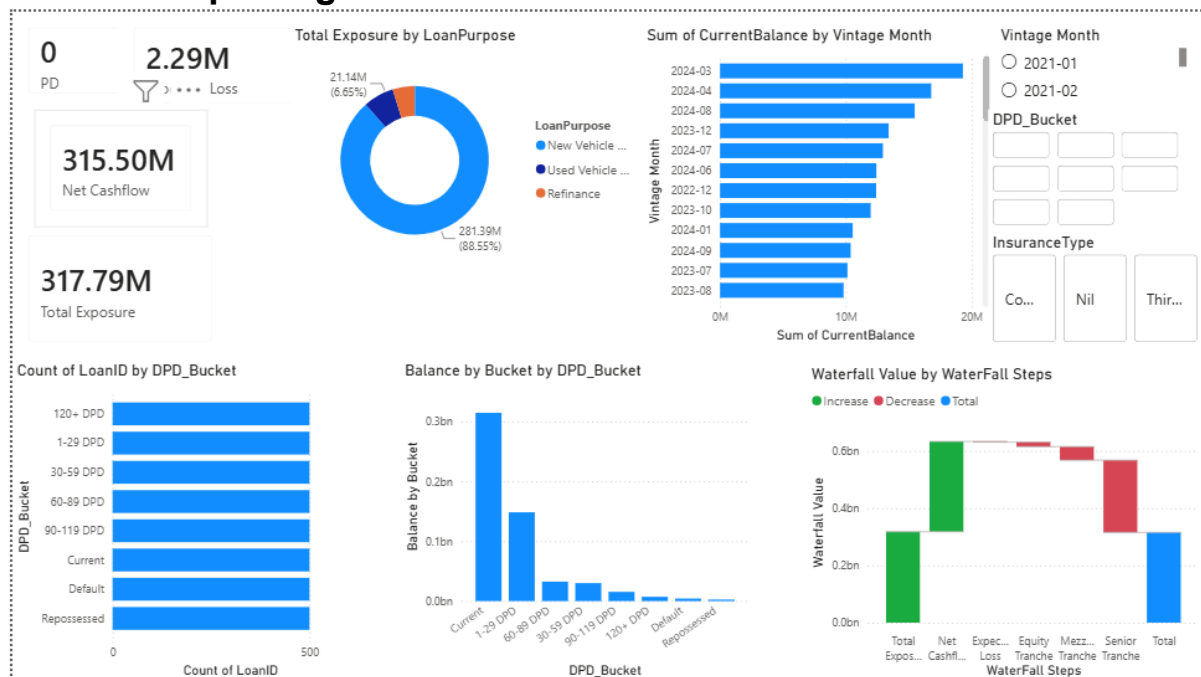
Vintage Trend

Line Chart

Investor Benefits

- Portfolio transparency
 - Risk monitoring
 - Better investment decisions
 - Regulatory reporting support
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Investor Reporting



Securitisation Risk Arena, Findings and Conclusion

Securitisation Risk Arena

The Risk Arena is a gamified analytics environment designed to simulate economic crisis scenarios.

Scenario 1

Mortgage Crisis

Expected increase in defaults and losses.

Scenario 2

Auto Loan Recession

Higher delinquency rates in vehicle loans.

Scenario 3

Interest Rate Shock

Borrowers face increased repayment burden.

Key Findings

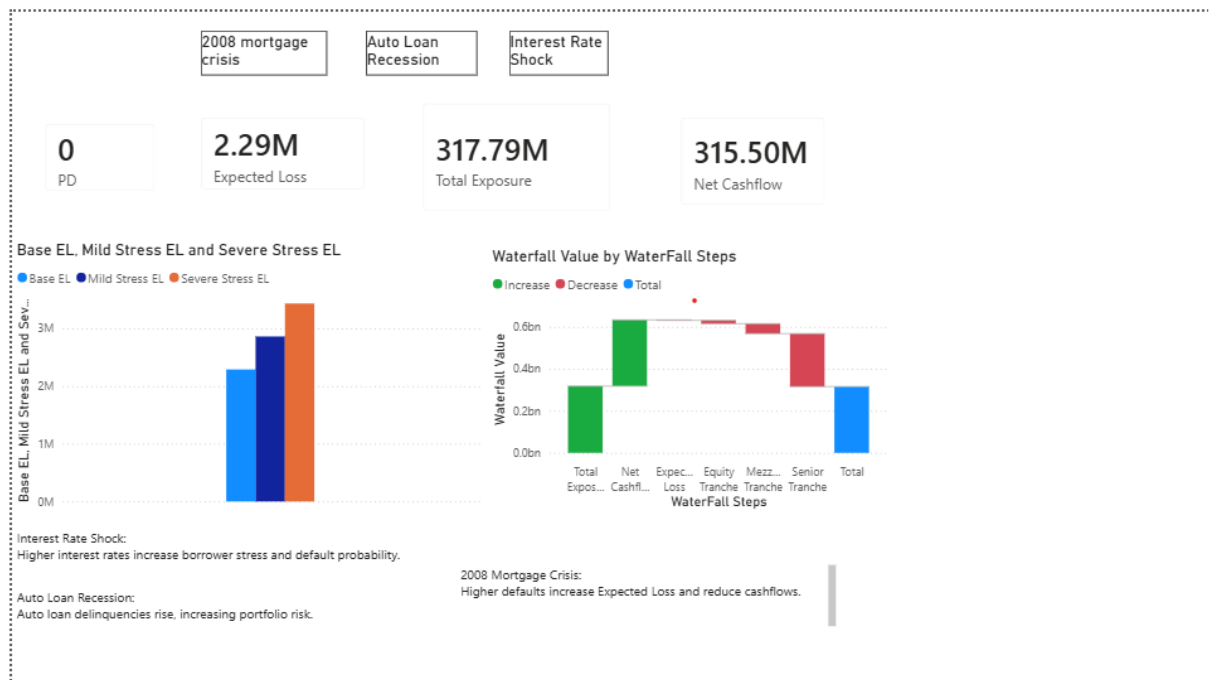
1. Portfolio risk increases significantly in higher DPD buckets.

2. Expected losses rise under stress conditions.
3. Older vintages exhibit higher delinquency rates.
4. Senior tranches remain relatively protected.
5. Equity tranches absorb the majority of losses.

Recommendations

1. Strengthen underwriting standards.
2. Monitor early-stage delinquencies.
3. Increase surveillance of high-risk vintages.
4. Perform quarterly stress testing.
5. Enhance investor reporting transparency.

Securitisation Arena



Conclusion

This project successfully developed an end-to-end securitisation risk analytics platform using Power BI. The solution integrates delinquency monitoring, IFRS 9 Expected Credit Loss modeling, vintage analysis, tranche cashflow waterfall analysis, stress testing, investor reporting, and crisis simulation capabilities.

The implemented dashboards provide actionable insights for risk managers, investors, and financial institutions. The project demonstrates how business intelligence and analytical modeling can improve portfolio monitoring, risk management, and decision-making within securitised asset portfolios.